Problem 26.7

A perpetuity pays $100 at the end of the first year. Each subsequent annual payment increases by $50. Calculate the present value at an annual effective interest rate of 10%.

Solution

The payments are

$$100, 150, 200, 250, \ldots$$

This can be written as a level perpetuity of 100 plus an increasing perpetuity with increments of 50:

$$100, 100, 100, \ldots$$

and

$$0, 50, 100, 150, \ldots$$

For a perpetuity-immediate with first payment P and annual increase Q, the present value is

$$PV = \frac{P}{i} + \frac{Q}{i^2}.$$

Here,

$$P = 100, \qquad Q = 50, \qquad i = 0.10.$$

Therefore,

$$PV = \frac{100}{0.10} + \frac{50}{(0.10)^2}.$$

Thus,

$$PV = 1000 + 5000 = 6000.$$

Therefore, the present value is

$$\boxed{6000}.$$